E-04-LEMMA-CM-threshold v1: small L3 threshold vs dissipation

ZFL Lab

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Enstrophy identity (where valid): $\frac{1}{2}\frac{d}{dt}\|\omega\|_2^2 = W - \nu\|\nabla\omega\|_2^2$, $W := \int \omega^T S\omega$.
Relation proven in CROSS-EF-05-LEMMA-bound: $|W(t)| \leq C\|u(t)\|_3\|\nabla\omega(t)\|_2^2$.
Assume $\sup_{t<T_*}\|u(t)\|_3 = M < \infty$.
Then $\frac{1}{2}\frac{d}{dt}\|\omega\|_2^2 \leq (CM - \nu)\|\nabla\omega\|_2^2$.
Threshold:

$$\boxed{CM < \nu \implies \frac{d}{dt}\|\omega(t)\|_2^2 \leq 0.}$$

Hence in this regime: $\|\omega(t)\|_2^2 \leq \|\omega_0\|_2^2$.
By E-03 (enstrophy bound =¿ continuation): $\boxed{\|u\|_3 < \nu/C \implies T_* = \infty.}$
Contrapositive (with same C from estimate): $\boxed{T_* < \infty \implies \sup_{t<T_*}\|u(t)\|_3 \geq \nu/C.}$
Distinction vs ESS:
ESS gives stronger: $T_* < \infty \implies \limsup_{t\to T_*}\|u(t)\|_3 = +\infty$.
This lemma gives small-data continuation via enstrophy: $\|u\|_3 < \nu/C \implies$ no blow-up.
It does NOT replace ESS. ESS is full necessity of C_{L^3}.
Classification: $\boxed{\text{THRESHOLD RELATION PROVEN: } CM < \nu}$
No new C_F. State remains: $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ in main, tag $508cbbf$ remains $\{C_E^{weak}\}$.